vector $\bar{a} = \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{bmatrix}$

$$\begin{bmatrix} a_1 & a_2 & \cdots & a_n \end{bmatrix}$$

matrix $\bar{A} = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & & \\ \vdots & & \ddots & \\ a_{m1} & & & a_{mn} \end{bmatrix}$

$m \times n$ matrix

$$\bar{A} = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$$

$m=2$ $\quad n=3$ $\quad 2 \times 3$

$m = n \rightarrow$ square matrix

$\bar{A}$ that $m \times n$

$\hookrightarrow \bar{A}^T$ is $n \times m$

$$\bar{A}^T = \begin{bmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{bmatrix}$$

$$X = \begin{bmatrix} 1 & 2 & 3 \end{bmatrix}$$

$$X^T = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$$

$$\bar{C} = \bar{A} + \bar{B} = \bar{B} + \bar{A}$$

$\hookrightarrow c_{ij} = a_{ij} + b_{ij}$

$$\bar{D} = \bar{A} - \bar{B}$$

$\hookrightarrow d_{ij} = a_{ij} - b_{ij}$

multiplication

* let $\bar{B} = \alpha \bar{A}$, α is scalar

$$b_{ij} = \alpha a_{ij}$$

* we can only multiply $m \times l$ with $l \times n$

$m \neq n$ is okay

$$\boxed{\bar{C} = \bar{A}\,\bar{B}}$$

$\hookrightarrow \quad c_{ij} = \sum_{k=1}^{l} a_{ik} b_{kj}$

ex: $\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} \begin{bmatrix} 7 & 8 \\ 9 & 10 \\ 11 & 12 \end{bmatrix}$

$2 \times \underline{3}$ $\quad$ $\underline{3} \times 2$

$m \times l$ $\quad$ $l \times n$

$m \times n$

$1 \times 7 + 2 \times 9 + 3 \times 11$

order matters :

x is $n \times \underline{1}$

$x^T x$ is scalar

$x x^T$ is $n \times n$

* $(AB)^T = B^T A^T$

* $x^T y = (x^T y)^T = y^T x$

$= \text{scalar}$

* Schur product

$C = A \circ B = B \circ A \rightarrow c_{ij} = a_{ij} b_{ij}$

Norms

$$\|x\|_1 = \sum_{i=1}^{n} |x_i| \quad , \quad (1\text{-norm})$$

$$\|x\|_2 = \sqrt{\sum_{i=1}^{n} x_i^2} \quad (2\text{-norm})$$

$$\|x\|_p = \left(\sum_{i=1}^{n} |x_i|^p\right)^{1/p} \quad (p\text{-norm})$$

for matrices:

$$\|A\|_{pp} = \|\mathrm{vec}(A)\|_p = \left(\sum_{i=1}^{m} \sum_{j=1}^{m} |a_{ij}|^p\right)^{1/p}$$

Frobenius - norm

$$||A||_F = \sqrt{\sum_{i=1}^{n} \sum_{j=1}^{m} a_{ij}^2} \quad \text{(2-norm of matrix)}$$

$$||A||_F = \sqrt{\operatorname{trace}(A^T A)}$$

$$\hookrightarrow \operatorname{trace}(B) = \sum_{i=1}^{n} b_{ii}$$

$$\boxed{\text{Linear independence}}$$

let $q_1, q_2, \cdots, q_m$ are $n \times 1$

let $\alpha_1, \alpha_2, \cdots, \alpha_m$ which are not all zero

linear combination

$$= \alpha_1 q_1 + \alpha_2 q_2 + \cdots + \alpha_m q_m$$

↳ $\sum_{i=1}^{m} \alpha_i q_i$

say we have

$$q_1 = [1, 0]$$

$$q_2 = [5, 0]$$

↳ $q_1 = 5 q_2$

therefore q_1 & q_2 are linearly dependent

$$\alpha_1 q_1 + \alpha_2 q_2 + \dots + \alpha_m q_m = 0$$

for some combo of α

which are not all zero

$$\hookrightarrow \quad q_1 = -\frac{\alpha_2}{\alpha_1} q_2 - \dots - \frac{\alpha_m}{\alpha_1} q_m$$

Say:

$$q_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \quad q_2 = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} \quad q_3 = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}$$

these are linearly independent

if have matrix $m > n$

then m vectors are lin. dep.

$\boxed{\text{Span}}$ $\text{span}\{q_1, q_2, \cdots, q_m\}$

is set of all linear combos of these vectors

Ex: $\text{span}\left\{ \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \end{pmatrix} \right\}$

$= R^2$

which defines a vector space

Let V be a vector space defined by above span.

$q_1, \cdots, q_m$ are basis for V if they are linearly independent

orthogonal vectors

if $q_i^T q_j = 0$ where $i \neq j$

$\boxed{\text{Rank}}$

null space :

~~#~~ dimensions null space of A is $n - \text{rank}(A)$

Special matrices

① square matrix $m=n$

② identity matrix $= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

such that $AI = IA = A$

③ if $m=n$ and $\text{rank}(A) = n$

then A is invertible

therefore null space is empty

if $\text{rank}(A) < n$ then A

is not invertible